

PRITAM SURYAWANSHI

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SUMMARY

M.Sc. Data Science student at Fergusson College with hands-on experience building and deploying production-grade machine learning systems across computer vision, forecasting, and analytics. Developed real-time AI solutions, including a real-time AI surveillance platform and a semiconductor trade-intelligence dashboard using Python, YOLOv8, FastAPI, and Streamlit. Seeking a Machine Learning / Data Science internship to build impactful real-world AI applications.

EDUCATION

M.Sc. Data Science — Fergusson College, Pune 2024 – 2027 (Expected)
B.Sc. Computer Science — PVG College, Pune Graduated 2024

SKILLS

Languages: Python, SQL
Machine Learning / Computer Vision: Scikit-learn, LightGBM, XGBoost, YOLOv8, TensorFlow, OpenCV
Deep Learning / Model Optimisation: ONNX, INT8 Quantisation, Hard Negative Mining
Time Series / Econometrics: ARIMA, VAR, Difference-in-Differences, Chow Test, Herfindahl-Hirschman Index (HHI)
Data Tools: Pandas, NumPy, Statsmodels, Jupyter Notebook, Plotly, Matplotlib, Seaborn
Deployment / MLOps: FastAPI, Streamlit, Git, MLflow, Kaggle

PROJECTS

Real-Time AI Surveillance System | [GitHub](#) | [Live Demo](#) Jan 2026 – Present

Tech: Python, YOLOv8, TensorFlow/Keras, OpenCV, ONNX, FastAPI

- Built an end-to-end **computer vision system** for detecting **weapons, smoke/fire, and violent activity** from images, videos, and live CCTV streams.
- Designed for real-world surveillance scenarios requiring low-latency, scalable, and deployable AI systems.
- Trained and optimised a YOLOv8 model achieving **Precision 93.7%, Recall 89.3%, mAP50 95.3%, and mAP50-95 69.3%**.
- Built an inference pipeline for real-time predictions across image, video, and live CCTV inputs.
- Compressed model size by 77% (25.9 MB to 6.2MB) using **ONNX** export and **INT8 quantisation** for faster deployment.

SemiTrack India – Semiconductor Import Intelligence Dashboard | [GitHub](#) | [Live Demo](#) 2025 – 2026

Tech: Python, Pandas, Statsmodels, ARIMA, VAR, Streamlit

- Built an interactive dashboard analysing **30 years (1995–2024)** of India's semiconductor import data.
- Identified long-term growth from **USD 0.24B to USD 23.05B (~10.2% annual growth)**.
- Analysed supply-chain concentration; the top 3 countries contributed **86.56% of imports**, highlighting geopolitical risk.
- Applied **ARIMA, VAR, Difference-in-Differences, and Chow Test** to forecast trends and evaluate policy impact.
- Built and deployed an interactive Streamlit dashboard for exploring trade trends, supplier concentration, and policy-driven forecast scenarios.

Customer Impulse Buying Prediction System | [GitHub](#) | [Live Demo](#) 2025

Tech: Python, LightGBM, Scikit-learn, Pandas, Streamlit

- Built a regression model to predict customer impulse buying scores (0–100) for e-commerce personalisation.
- Performed feature engineering using demographic, behavioural, and purchase data.
- Achieved **R² = 0.85, RMSE 8–9, MAE 6–7** using LightGBM.
- Developed a Streamlit app for real-time predictions with feature-importance-based interpretation.

CERTIFICATIONS

- NVIDIA Deep Learning Institute — [Deep Learning Fundamentals](#)
- Machine Learning in Data Science — [Great Learning](#)